

CHEMICAL CHANGES – EXTRACTION OF METALS

KEY VOCABULARY

Ore: a rock containing enough metal to extract profitably.

Reduction: loss of oxygen (or gain of electrons).

Reduction by carbon: used for metals less reactive than carbon.

Electrolysis: used to extract metals more reactive than carbon.

COMMON MISCONCEPTIONS

~~— All metals are extracted by carbon.~~

✓ **Reactive metals (above carbon) need electrolysis.**

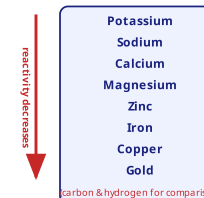
~~— Gold must be extracted from an ore.~~

✓ **Gold is unreactive and found as the metal itself.**

~~— Reduction means adding oxygen.~~

✓ **Reduction is the LOSS of oxygen.**

EXTRACTION METHOD



Above carbon → **electrolysis**. Below carbon → **reduction by carbon**.

1 DEFINE REDUCTION

State what is meant by **reduction** in terms of oxygen. **[1 mark]**

2 CHOOSE A METHOD

Explain why aluminium is extracted by electrolysis, not by carbon. **[2 marks]**

3 IRON FROM IRON OXIDE

Carbon reduces iron oxide. **Identify** which substance is reduced. **[1 mark]**

4 WHY ELECTROLYSIS COSTS

Explain why extracting metals by electrolysis is expensive. **[2 marks]**

5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

Metals ____ carbon in the series are extracted by ____.

carbon

electrolysis

above

below

6 BIOLEACHING

Suggest one advantage of using bacteria (bioleaching) to extract copper. **[1 mark]**